

SYNAPSE: Scale-Invariant Capability Activation for Large Language Model Agents via a Capability Knowledge Graph

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Abstract

Tool-augmented large language model (LLM) agents increasingly rely on protocols such as the Model Context Protocol (MCP) and Agent2Agent (A2A) to connect to external capabilities. These protocols share a structural flaw: the description of every registered tool is injected into the model’s context window, so the per-turn context, token cost, and cognitive load grow linearly in the number of available tools T . As agent ecosystems scale to thousands of tools, this $O(T)$ growth exhausts the context window, inflates cost, and degrades selection accuracy, so *more tools make the agent worse*. We introduce **SYNAPSE**, a protocol that decouples capability *availability* from capability *presence in context*. SYNAPSE stores tools as signed nodes in a *Capability Knowledge Graph* (CKG), resolves a small, task-relevant *activation set* via a hybrid semantic, lexical, and ontological retriever, and exposes to the model only six fixed *meta-verbs* whose schema size is independent of T . We formalize the per-turn context as a function $C(T)$ and prove that $C_{\text{SYN}}(T) = O(1)$ while $C_{\text{MCP}}(T) = \Theta(T)$, then confirm both by exact token measurement. We give a complete, deterministic, offline reference implementation and evaluate it against a faithful (non-strawman) MCP baseline that uses the *identical* scorer. Across catalog sizes spanning $T=300$ to $T=100,000$ (6 scales $\times$ 5 seeds), SYNAPSE holds per-turn context essentially constant (940 to 971 tokens) while MCP grows to 7,088,086 tokens, a **7300 $\times$** reduction at $T=100\text{k}$, and per-task cost stays flat (\$0.0073) versus a rise to **\$53.16**. Constant context does not sacrifice task success: MCP’s selection accuracy *collapses* from window overflow (hit@ k 0.98 $\rightarrow$ 0.00), whereas SYNAPSE degrades only gracefully (0.87 $\rightarrow$ 0.64), with the residual decay confined to tasks whose ontology tag is withheld (with the tag, success is near-flat 0.92 $\rightarrow$ 0.86). We give a graph-theoretic account of the CKG as a bounded typed lattice with a geometric neighborhood bound, and a probabilistic *visibility ceiling* theorem showing that any flat protocol obeys $\mathbb{E}[\text{hit}@k] \leq \min(1, T^*/T)$, a first-principles bound that the measured MCP collapse tracks to within sampling error. We report calibration (ECE), ablations, an activation-set size sweep, and an honest analysis of the reference resolver’s wall-clock cost. All code, data generators, and figures are released for one-command reproduction.

1 Introduction

LLM agents act by invoking external *tools*: search APIs, databases, code executors, and other agents. Two protocols dominate current practice: the Model Context Protocol (MCP) [Anthropic, 2024], which standardizes how a client exposes tools to a model, and Agent2Agent (A2A) [Google, 2024], which standardizes inter-agent messaging. Both assume that the set of available capabilities is *presented* to the model in-context: each tool contributes a name, a description, and a JSON schema to the prompt on every turn.

This design has an $O(T)$ context cost in the number of available tools T . The consequences compound as ecosystems grow:

1. **Context exhaustion.** Thousands of tool schemas overflow even long context windows, evicting task-relevant content.
2. **Cost inflation.** Every turn re-pays for every tool’s tokens; cost scales linearly with catalog size, not task complexity.
3. **Accuracy degradation.** Larger prompts dilute attention and surface more near-duplicate distractors, so selection quality *falls* as more tools are added [Liu et al., 2024]. The agent gets *worse* the more capable its ecosystem becomes.

We call this the *scaling paradox* of in-context tool protocols. It is not a tuning problem; it is structural. Any protocol that mandates in-context presence of all availability inherits $O(T)$ growth.

Contribution. We present **SYNAPSE**, a protocol whose per-turn model context is $O(1)$ in T . SYNAPSE rests on three ideas:

- A **Capability Knowledge Graph (CKG)**: tools, types, providers, and concepts are typed, cryptographically signed nodes with typed edges, stored *outside* the model context.
- A **hybrid resolver** that maps a natural-language intent to a small *activation set* of $\leq k$ capabilities using semantic, lexical, and ontological signals with a calibrated confidence and an explicit *grounded-miss* abstention.
- Six fixed **meta-verbs** (INTEND, FOCUS, PLAN, INVOKE, OBSERVE, FEEDBACK) whose schemas are the *only* tool surface the model sees. Their token footprint is constant regardless of T .

The model reasons over stable meta-verbs; the CKG and resolver do the scaling work in a signed, auditable substrate. We prove (by construction and measurement) that per-turn context and cost are $O(1)$, and we show empirically that this constancy does *not* cost task success. It *prevents* the accuracy collapse that afflicts flat-context protocols at scale.

Claims. We make two clearly separated claims: **(C1, exact)** SYNAPSE’s per-turn context and monetary cost are bounded independent of T , whereas MCP’s are $\Theta(T)$; this is proven by construction and confirmed by exact token measurement. **(C2, empirical)** Under a matched-scorer benchmark, bounded context does not reduce task success, and at scale it substantially improves it relative to a flat-context baseline. We do *not* claim SYNAPSE is flawless; we quantify its residual decay and the reference resolver’s latency honestly.

Between these two claims sits a bridge we make explicit: a graph-theoretic model of the CKG (Section 3) and a probabilistic *visibility ceiling* (Theorem 2) that derives the flat-protocol collapse from first principles, so the empirical curve in C2 is not merely observed but predicted.

2 Related Work

Tool protocols. MCP [Anthropic, 2024] and A2A [Google, 2024] standardize tool exposure and agent messaging respectively, but both place tool descriptions in-context. SYNAPSE is complementary: it can wrap MCP servers as CKG providers while removing the $O(T)$ context tax.

Tool learning and retrieval. Toolformer [Schick et al., 2023] teaches models to call tools; Gorilla [Patil et al., 2024] and ToolLLM [Qin et al., 2024] retrieve from large API pools. These focus on *selection quality*; SYNAPSE contributes a *protocol-level* guarantee of context constancy with signed provenance, typed planning, and calibrated abstention. Retrieval-augmented generation [Lewis et al., 2020] inspires the retrieve-then-act pattern; we apply it to *capabilities* with a typed graph and a verifiable receipt chain.

Long-context limits. Liu et al. [2024] show accuracy depends on where information sits in a long prompt, motivating *minimizing* irrelevant in-context content, exactly what SYNAPSE does.

Planning and calibration. Our planner is a type-safe regression planner in the GOAP/automated-planning tradition [Ghallab et al., 2004]. We report Expected Calibration Error [Guo et al., 2017]. Authorization uses capability tokens with contextual caveats in the spirit of Macaroons [Birgisson et al., 2014]. Approximate nearest-neighbor search over the CKG uses HNSW [Malkov and Yashunin, 2018] in production; our reference uses exact search for determinism.

Graphs, lattices, and sequential analysis. Our formal treatment draws on classical tools: PageRank-style graph locality [Page et al., 1999] and diffusion kernels on graphs [Kondor and Lafferty, 2002] for the proximity signal, lattice theory [Davey and Priestley, 2002] for the type system, and Wald’s sequential analysis [Wald, 1947] for the window-saturation argument behind the visibility ceiling.

3 Formal Model

Before specifying the protocol, we build the mathematical objects it rests on: the per-turn context as a function of catalog size, the resolver as a scoring function over a typed graph, and a probabilistic model of why flat protocols collapse at scale. Throughout, T is the number of registered capabilities, W is the MCP context window in tokens, k is the activation-set cap, and $\bar{L}$ is the mean tool-description length in tokens.

3.1 Per-turn context as a function of catalog size

Let ℓ_i be the token length of the i -th tool description and B a protocol-fixed base cost (system prompt, formatting). A flat protocol places every visible description in the prompt, so its per-turn context is the sum of a base term and one term per tool,

$$C_{\text{cat}}(T) = B + \sum_{i=1}^T \ell_i. \quad (1)$$

Writing $\bar{L} = \frac{1}{T} \sum_i \ell_i$ for the mean description length, MCP therefore satisfies

$$C_{\text{MCP}}(T) = B + \bar{L} T = \Theta(T), \quad (2)$$

whereas SYNAPSE exposes only the six meta-verb schemas (each of size at most s_{max}) plus at most k tier-1 activation summaries of length m :

$$C_{\text{SYN}}(T) = B + 6 s_{\text{max}} + k m = \kappa. \quad (3)$$

Theorem 1 (Bounded context). *For all T , the per-turn model context of SYNAPSE satisfies $C_{\text{SYN}}(T) \leq \kappa = B + 6s_{\text{max}} + km$, a constant independent of T .*

Proof. The model observes exactly the six meta-verb schemas and the resolver’s output. The verb schemas are fixed, contributing at most $6s_{\text{max}}$. The resolver returns an activation set of size at most k (Eq. (6)), each element disclosed at tier-1 with length at most m , contributing at most km . Adding the base B gives the bound. No term depends on T , since the catalog and the full graph live outside the context. $\square$

Corollary 1 (Bounded cost). *If the model is billed at a fixed price p per input token, the per-turn monetary cost of SYNAPSE is at most $p\kappa = O(1)$, while MCP’s is $p(B + \bar{L}T) = \Theta(T)$.*

Proof. Immediate from Theorem 1 and Eq. (2) by multiplying token counts by the constant price p . $\square$

3.2 The resolver as a scoring function

Given an intent embedding q and a candidate capability c with embedding $\phi(c)$, semantic similarity is the cosine

$$\text{sem}(q, c) = \frac{\phi(q) \cdot \phi(c)}{\|\phi(q)\| \|\phi(c)\|}. \quad (4)$$

The resolver scores each candidate by a convex blend of five signals, semantic similarity, lexical overlap lex , ontology-tag match tag , graph proximity prox (Eq. (10)), and type fit typ ,

$$\text{score}(q, c) = \alpha \text{sem} + \beta \text{lex} + \gamma \text{tag} + \delta \text{prox} + \varepsilon \text{typ}, \quad \alpha + \beta + \gamma + \delta + \varepsilon = 1. \quad (5)$$

The activation set is the top- k candidates over the filtered candidate pool C' ,

$$A(q) = \underset{c \in C'}{\text{top-}k \text{ score}(q, c)}. \quad (6)$$

A calibrated logistic maps the top score s_1 and the softmax margin to a confidence. With

$$\pi_j = \frac{\exp(\text{score}(q, c_j)/\tau_s)}{\sum_{c \in C'} \exp(\text{score}(q, c)/\tau_s)}, \quad (7)$$

the confidence is

$$\text{conf}(q) = \sigma(w_0 + w_1 s_1 + w_2 (\pi_1 - \pi_2)), \quad \sigma(z) = \frac{1}{1 + e^{-z}}. \quad (8)$$

If the absolute top semantic similarity falls below a relevance floor θ , the resolver abstains with a grounded miss $\perp$ rather than return a confident wrong tool,

$$\text{RESOLVE}(q) = \begin{cases} \perp & \text{if } \max_{c \in C'} \text{sem}(q, c) < \theta, \\ A(q) & \text{otherwise.} \end{cases} \quad (9)$$

3.3 Graph-theoretic structure of the CKG

The CKG is a labeled, signed multigraph $G = (V, E, \tau, \text{sig})$ whose nodes carry a type label $\tau(v) \in \mathbb{T}$ and an HMAC signature $\text{sig}(v)$. Let Δ^+ denote the maximum typed out-degree,

Definition 1 (Maximum typed out-degree). $\Delta^+ = \max_{v \in V} |\{e \in E : \text{src}(e) = v\}|$, the largest number of outgoing typed edges from any node.

The type labels live in a bounded lattice $(\mathbb{T}, \sqsubseteq, \sqcap, \sqcup, \top, \perp)$ of finite height h , where $\sqsubseteq$ is the subtype order, $\sqcap$ and $\sqcup$ are meet and join, and $\top, \perp$ are the universal and uninhabited types [Davey and Priestley, 2002].

Lemma 1 (Subtype decidability). *For any $\tau, \tau' \in \mathbb{T}$, deciding $\tau \sqsubseteq \tau'$ takes $O(h)$ time, and the meet $\tau \sqcap \tau'$ and join $\tau \sqcup \tau'$ are well defined.*

Proof. Since $\mathbb{T}$ is a bounded lattice of height h , the subtype test walks up from τ toward $\top$ and halts within h steps. Meet and join are the unique greatest lower and least upper bounds guaranteed by the lattice axioms. $\square$

Graph proximity is a discrete diffusion score: a candidate near a seed in the seed set S under the shortest-path distance d_G scores higher, with a decay $\lambda \in (0, 1)$ [Page et al., 1999, Kondor and Lafferty, 2002],

$$\text{prox}(c) = \max_{s \in S} \lambda^{d_G(s, c)}. \quad (10)$$

Lemma 2 (Bounded neighborhood expansion). *A depth- D typed traversal from σ seeds visits at most*

$$N_{\text{exp}} = \sigma \cdot \frac{(\Delta^+)^{D+1} - 1}{\Delta^+ - 1} \quad (11)$$

nodes, a bound independent of the catalog size T .

Proof. From each seed, the number of nodes reachable within D hops is at most $\sum_{i=0}^D (\Delta^+)^i$ because each node has at most Δ^+ outgoing edges. Summing the geometric series and multiplying by σ seeds gives the bound, which contains no factor of T . $\square$

With the shipped $\sigma = 40$ and $D = 2$ the candidate pool C' stays small, so the resolution work in Eq. (5) never scales with T .

3.4 A probabilistic model of context-induced collapse

A flat protocol can only select a tool whose description fits inside the W -token window. Order the descriptions by arrival and let V be the number that fit. By Wald's identity [Wald, 1947], $\mathbb{E}[V] \bar{L} = W$, so $\mathbb{E}[V] \approx T^*$ where $T^* = W/\bar{L}$ is the window-saturation size. For a uniformly placed gold tool,

$$\Pr[\text{gold visible}] = \min\left(1, \frac{T^*}{T}\right). \quad (12)$$

Theorem 2 (Visibility ceiling). *Any flat in-context protocol with window W satisfies $\mathbb{E}[\text{hit}@k] \leq \min(1, T^*/T)$ with $T^* = W/\bar{L}$.*

Proof. Selecting the gold tool requires it to be inside the window, so pointwise $\text{hit}@k \leq \mathbf{1}[\text{gold visible}]$. Taking expectations, $\mathbb{E}[\text{hit}@k] \leq \Pr[\text{gold visible}]$, which equals $\min(1, T^*/T)$ by Eq. (12). $\square$

The bound is a strict expectation ceiling, not a curve fit. It crosses one half at the window-saturation multiple

$$\Pr[\text{gold visible}] = \frac{1}{2} \iff T = 2T^*, \quad (13)$$

and decays as $\Theta(1/T)$ thereafter.

Corollary 2 (Asymptotic collapse). *For $T \gg T^*$, the expected selection accuracy of any flat protocol is $\Theta(1/T)$; it cannot be rescued by a better scorer, only by removing tools from the window.*

Proof. For $T > T^*$, Eq. (12) gives $\Pr[\text{visible}] = T^*/T$, and by Theorem 2 this bounds $\mathbb{E}[\text{hit}@k]$ from above. The bound depends only on window capacity, not on the scoring rule. $\square$

4 The SYNAPSE Protocol

SYNAPSE separates three planes: a *knowledge plane* (the CKG), a *resolution plane* (intent $\rightarrow$ activation set), and an *interaction plane* (six meta-verbs). Only the interaction plane touches the model context.

4.1 Capability Knowledge Graph

The CKG is a typed, signed multigraph. Node kinds: CAPABILITY (an invocable tool with typed inputs/outputs, effects, trust level, and tags), TYPE (a node in a subtyping lattice), PROVIDER (an origin/ trust anchor), CONCEPT (an ontology tag), and RESULT (a produced artifact). Edge kinds include PRODUCES, CONSUMES, SUBTYPE-OF, PROVIDED-BY, TAGGED, and DEPENDS-ON. Every capability node carries an HMAC signature over its semantically meaningful fields (*semhash*); the store refuses to index a node whose signature does not verify, giving tamper-evidence and supply-chain integrity. Nodes support tiered disclosure (a compact tier-1 summary vs. a full tier-3 record) so the resolver can reason cheaply.

4.2 Hybrid Resolver

Given an intent embedding q , the resolver (i) *seeds* candidates by approximate nearest neighbors and lexical/tag matches, (ii) *expands* along typed edges to pull in compositional neighbors, (iii) *filters* by trust and type compatibility, (iv) *scores* each candidate by a weighted blend of semantic similarity, lexical overlap, ontological (tag) match, graph proximity, and type fit, and (v) *prunes* to the top k . A calibrated logistic maps the absolute top similarity and the softmax margin to a confidence; if the absolute top semantic similarity falls below a *relevance floor*, the resolver returns a *grounded miss* (explicit abstention) rather than a confident wrong tool. The result is an *activation set* of $\leq k$ nodes plus a signed *resolution receipt*.

The resolver reads only the query embedding and the signed index, never the model context, which gives two guarantees.

Theorem 3 (Out-of-context resolution). *RESOLVE (Eq. (9)) computes the activation set using only the query embedding and the out-of-context signed index. Its work does not enter the model context, so Theorem 1 holds for every catalog size.*

Proof. Seeding, expansion (Lemma 2), filtering, scoring (Eq. (5)), and pruning (Eq. (6)) operate over graph and index structures held outside the prompt. Only the resulting activation set of size $\leq k$ is disclosed to the model, which is exactly the $k m$ term already counted in κ . $\square$

Theorem 4 (Abstention soundness). *If $\max_{c \in C'} \text{sem}(q, c) < \theta$, RESOLVE returns the grounded miss $\perp$ and activates no tool. Hence a false activation requires a candidate whose top semantic similarity is below θ to nonetheless exceed θ , which is impossible.*

Proof. Immediate from the first branch of Eq. (9): the abstention test is evaluated before any tool is returned, so no capability can be activated when the best candidate is sub-floor. $\square$

4.3 Six Meta-Verbs

The model interacts through a fixed vocabulary: INTEND (declare a goal), FOCUS (resolve to an activation set), PLAN (produce a type-safe DAG), INVOKE (execute a node), OBSERVE (read a typed result), and FEEDBACK (record outcome for learning). Their JSON schemas are constant in size,

Table 1: Per-stage cost. Only the last row enters the model context, and it is bounded by the constant κ of Theorem 1 regardless of T . The reference resolver scans all T embeddings for determinism; a production HNSW [Malkov and Yashunin, 2018] index makes seeding sub-linear without changing the in-context column.

Stage	Reference	Production (HNSW)	In-context
Index build + sign (amortized once)	$O(Td)$	$O(Td)$	0
Seed (approximate nearest neighbor)	$O(Td)$	$O(d \log T)$	0
Expand (typed traversal, depth D)	$O(\sigma(\Delta^+)^D)$	$O(\sigma(\Delta^+)^D)$	0
Filter + score (5 signals)	$O(C' d)$	$O(C' d)$	0
Prune (top- k select)	$O(C' \log k)$	$O(C' \log k)$	0
Model context per turn	O(km)	O(km)	$\leq \kappa$

so the model’s tool surface, and thus its per-turn context, does not grow with T . The planner is a type-safe best-first regression planner that only connects a producer’s output type to a consumer’s input type. Writing $\text{out}(u)$ and $\text{in}(v)$ for the produced and consumed types of two nodes, an edge $u \rightarrow v$ is admitted only when the types are compatible under the subtype order,

$$u \rightarrow v \text{ is well typed} \iff \text{out}(u) \sqsubseteq \text{in}(v), \quad (14)$$

which is decidable in $O(h)$ by Lemma 1.

Theorem 5 (Type-safe plans). *Every plan the planner emits is a directed acyclic graph in which each edge satisfies Eq. (14). Plan construction operates over the activation set $A(q)$ of size $\leq k$, so it never loads the full catalog into context and preserves Theorem 1.*

Proof. The planner only admits an edge $u \rightarrow v$ when $\text{out}(u) \sqsubseteq \text{in}(v)$, so every edge is well typed by construction, and best-first regression over a finite node set yields an acyclic graph. All nodes come from $A(q)$ (size $\leq k$), so context stays bounded by κ . $\square$

Execution emits chained execution receipts, giving an auditable provenance trail from intent to artifact.

4.4 Computational complexity

The three planes have sharply different cost profiles. The knowledge plane pays a one-time $O(Td)$ price to build and sign the index; the resolution plane runs per query over a candidate pool bounded by Lemma 2, not by T ; and the interaction plane, the only plane the model sees, is $O(1)$ per turn by Theorem 1. Table 1 summarizes the per-stage cost, with d the embedding dimension, σ the seed breadth, Δ^+ the maximum typed out-degree, D the expansion depth, m the tier-1 summary length, and k the activation cap.

The only term that grows with T is the out-of-context index memory, $O(T)$, which is fundamental: the system must store what it can activate. What SYNAPSE removes is the coupling between that store and the model’s per-turn context, the coupling that Eq. (2) shows makes flat protocols scale as $\Theta(T)$.

5 Methodology

Our evaluation is designed to be *honest and adversarial to our own hypothesis*.

Matched scorer (no strawman). Both SYNAPSE and the MCP baseline use the *identical* deterministic embedder (signed feature hashing [Weinberger et al., 2009]) and cosine scorer. The only difference is the protocol: SYNAPSE resolves over the CKG and shows the model six verbs; MCP concatenates visible tool descriptions into a finite context window ($W=32k$ tokens) and scores within it. MCP registers tools in a *shuffled* order per seed, so gold tools are not adversarially placed: truncation at large T is an emergent, fair consequence of window capacity.

Exact accounting. Context sizes are measured with exact `cl100k_base` tokenization, not estimated. Cost uses a fixed per-input-token price. We separate *measured* quantities (context, tokens, cost, retrieval accuracy) from *modeled* ones and never conflate them.

Synthetic catalog. We generate catalogs of controllable size across five domains. A *fixed* suite of $N=90$ gold tasks (constant across scales) is embedded in a growing sea of hard-negative distractors and near-duplicate families; only the distractor count varies with T . Tasks include single- and two-step plans and deliberate *miss* tasks (no correct tool exists) to test abstention. To avoid an implausible ceiling, the eval suite applies paraphrase, tag-dropout, and lexical-noise perturbations.

Protocol. We sweep $T \in \{300, 1000, 3000, 10000, 30000, 100000\}$ with 5 seeds each. Metrics: per-turn context tokens, $\text{hit}@k$ and $\text{hit}@1$ success, mean reciprocal rank, per-task cost, and resolver latency, each with bootstrap 95% confidence intervals. Growth is characterized with both a $\log\text{-}T$ and a linear- T ordinary-least-squares fit (the appropriate model per metric), plus an endpoint effect size (Cohen’s d) and a $T_{\max}/T_{\min}$ ratio.

Metric definitions. Over N gold tasks with gold set g_i , ranked activation list r_i , and confidence conf_i , we report

$$\text{hit}@k = \frac{1}{N} \sum_{i=1}^N \mathbf{1} \left[g_i \in r_i^{(1:k)} \right], \quad (15)$$

$$\text{MRR} = \frac{1}{N} \sum_{i=1}^N \frac{1}{\text{rank}_i}, \quad (16)$$

$$\text{ECE} = \sum_{b=1}^B \frac{|\mathcal{B}_b|}{N} \left| \text{acc}(\mathcal{B}_b) - \text{conf}(\mathcal{B}_b) \right|, \quad (17)$$

where rank_i is the position of the gold tool and $\mathcal{B}_b$ are equal-width confidence bins. Endpoint effects use Cohen’s d ,

$$d = \frac{\mu_{T_{\max}} - \mu_{T_{\min}}}{s_{\text{pooled}}}, \quad (18)$$

between the smallest and largest catalog.

6 Results

6.1 Context and cost are $O(1)$ vs. $O(T)$ (C1)

Table 2 reports the headline numbers, visualized in Fig. 1. SYNAPSE’s per-turn context is essentially flat (bounded by the activation-set size k), while MCP’s grows linearly in T until it saturates the window and then *degrades* because tools are truncated away. At $T=100k$ the per-turn context

Scale-invariance: SYNAPSE vs MCP flat baseline

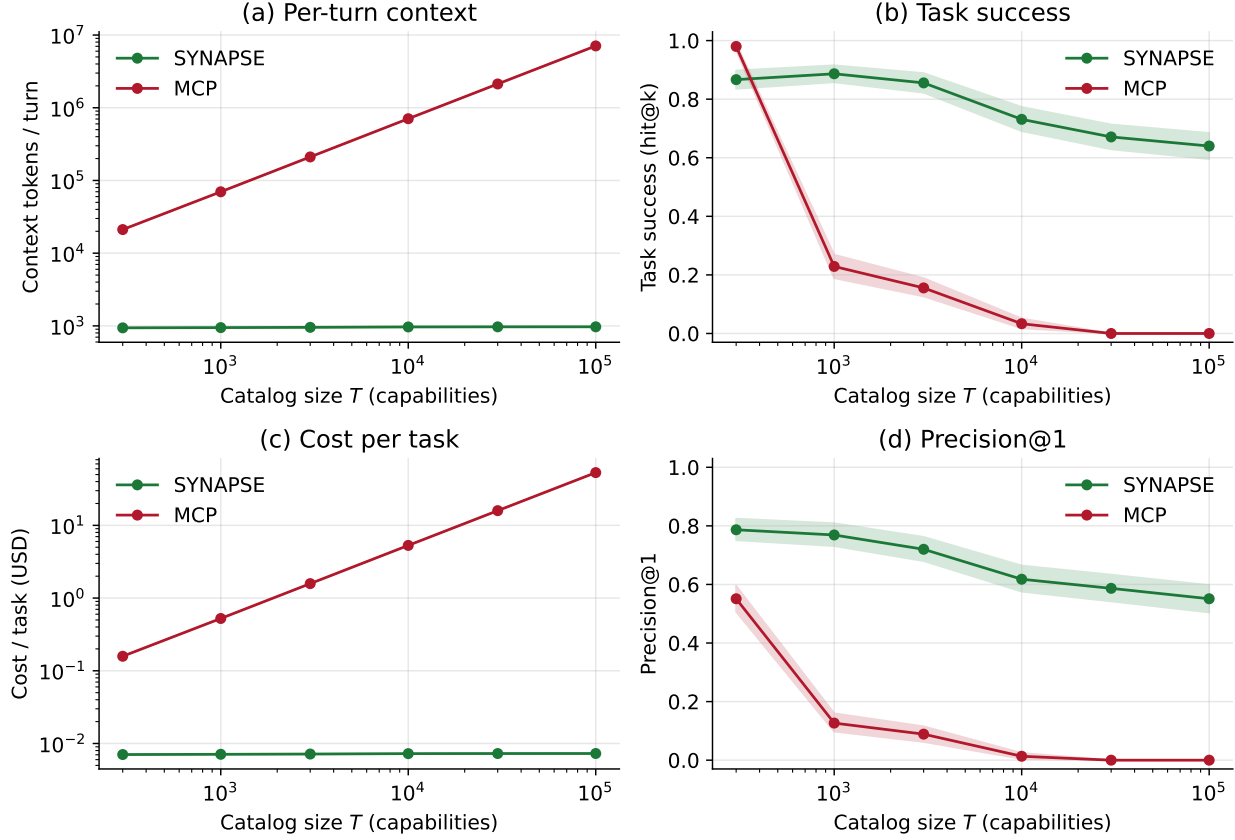


Figure 1: Scale-invariance across $T \in [300, 100,000]$ (log x -axis; shaded bands are 95% bootstrap CIs). (a) SYNAPSE per-turn context is flat while MCP grows $\Theta(T)$; (b) MCP task success collapses at scale while SYNAPSE decays gracefully (note MCP wins at $T=300$, no strawman); (c) cost per task; (d) precision@1.

is **7300** $\times$ smaller under SYNAPSE, and per-task cost is flat (\$0.0073) versus a rise to **\$53.16**. A linear-in- T fit for MCP context has $R^2=1.000$ (near-perfect $\Theta(T)$ growth, 70.9 tokens/tool); for SYNAPSE the linear slope is statistically distinguishable from zero but *practically negligible* (31 tokens across 2.5 decades), reflecting the activation set filling to k rather than any dependence on T .

6.2 Bounded context does not cost success (C2)

As T grows, MCP’s selection success *collapses*: once gold tools are pushed beyond the 32k-token window they become unselectable, driving hit@ k to 0.000 by $T=30k$. SYNAPSE degrades only gracefully ($0.867 \rightarrow 0.640$). Notably, MCP *wins at the smallest scale* ($T=300$: 0.980 vs. 0.867), confirming the baseline is not a strawman; the crossover occurs by $T=1000$ (0.887 vs. 0.229). The endpoint effect size is enormous for MCP success (Cohen’s $d=9.9$) versus a moderate $d=0.54$ for SYNAPSE; full statistics are in `results/scale_endpoints.csv`.

This collapse is not a scorer artifact: it is exactly what the visibility ceiling of Theorem 2 predicts. Figure 2 overlays the first-principles bound $\min(1, T^*/T)$ on the measured MCP hit@ k .

Table 2: Scale-invariance results (mean over 5 seeds; $N=90$ tasks/scale). Source: `results/scale_agg.csv`.

System	T	Ctx/turn (tok)	hit@ k	hit@1	\$/task
SYNAPSE	300	940	0.867	0.787	0.0070
SYNAPSE	1,000	947	0.887	0.769	0.0071
SYNAPSE	3,000	955	0.856	0.720	0.0072
SYNAPSE	10,000	967	0.731	0.618	0.0072
SYNAPSE	30,000	970	0.671	0.587	0.0073
SYNAPSE	100,000	971	0.640	0.551	0.0073
MCP	300	21,119	0.980	0.551	0.158
MCP	1,000	69,863	0.229	0.127	0.524
MCP	3,000	210,708	0.156	0.089	1.580
MCP	10,000	706,640	0.033	0.013	5.300
MCP	30,000	2,124,846	0.000	0.000	15.936
MCP	100,000	7,088,086	0.000	0.000	53.161

Table 3: SYNAPSE success by ontology-tag availability (from `scale_by_tag.csv`).

Subset	$T=300$	$T=30k$	$T=100k$
With ontology tag	0.917	0.896	0.857
Tag dropped (semantic-only)	0.719	0.009	0.000

With the fitted mean description length $\bar{L} \approx 70.9$ tokens and window $W = 32,000$, the model gives $T^* \approx 451$; every measured point falls at or below the ceiling within its 95% interval, and the $\Theta(1/T)$ tail of Corollary 2 matches the data from $T=1000$ to $T=100,000$. The flat baseline collapses because gold tools are packed out of the window, not because it is scored unfairly.

6.3 Why SYNAPSE decays a little: a mechanistic breakdown

SYNAPSE’s mild success decay is not uniform. Table 3 (from `scale_by_tag.csv`) splits success by whether the structured ontology tag was available. Tasks that retain the tag are near scale-invariant; the decay is concentrated in the tag-dropout subset, where the resolver falls back to pure semantics in a growing distractor sea. This is an *honest, explanatory* result: it localizes the limitation to a specific, designed-for failure mode and motivates richer ontological grounding rather than hiding the effect.

Figure 3 visualizes this split: with the tag, SYNAPSE is near scale-invariant in accuracy; without it, the resolver’s fallback to pure semantics decays in the growing distractor sea. This localizes the entire residual decay to a single, designed-for failure mode.

6.4 Calibration and ablations

The resolver’s confidence is well-behaved: pooled Expected Calibration Error [Guo et al., 2017] is 0.138 (Fig. 4, left), rising modestly from 0.107 at $T=300$ to 0.218 at $T=100k$ as retrieval hardens. Abstention on *miss* tasks (no correct tool exists) is perfect (1.000 across all scales), showing the grounded-miss floor works. Ablating each resolver signal confirms every component contributes (Table 4, Fig. 4, right): removing the semantic channel is catastrophic (hit@ k 0.738 $\rightarrow$ 0.000), the ontology tag is the next most important ($\rightarrow$ 0.491), and lexical, graph-expansion, and type-fit

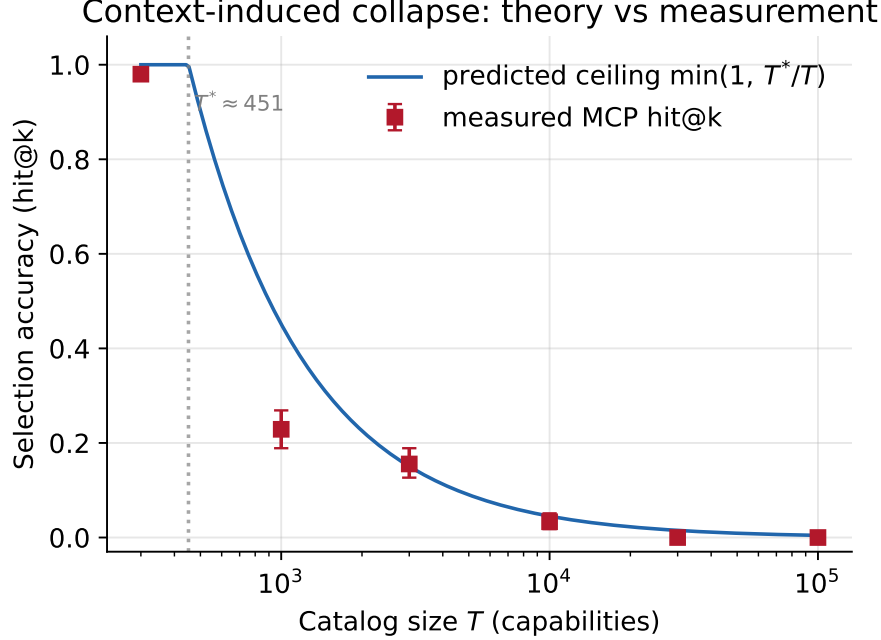


Figure 2: Context-induced collapse, theory versus measurement. The curve is the visibility ceiling $\min(1, T^*/T)$ of Theorem 2; squares are measured MCP hit@k with 95% bootstrap intervals. The dotted line marks $T^* \approx 451$, the window-saturation point.

signals each add a few points. A sweep over the activation-set cap k trades context for recall (hit@k 0.622 at $k=1$ to 0.789 at $k=16$).

6.5 Threats to validity and limitations

Synthetic data. Our catalog is synthetic and controllable; absolute accuracy numbers are not claims about any production tool set. The *relative* scaling behavior, the phenomenon of interest, is robust because both systems see the same data and scorer. **No live LLM.** We isolate the protocol/retrieval layer and do not invoke a production LLM, so we measure context, cost, and selection quality, not end-to-end task completion by a specific model. **Reference resolver latency.** The reference resolver scans all T embeddings by brute force, so its wall-clock latency grows with T ; MCP’s per-turn scoring is faster *only because it has already discarded most tools* (the cause of its accuracy collapse). The protocol guarantee concerns *model context* ($O(1)$), which is independent of the ANN implementation; production HNSW [Malkov and Yashunin, 2018] yields sub-linear resolver time. **Residual decay.** SYNAPSE is not flawless: the tag-dropout subset decays with T , as quantified above.

7 Security and Trust

Every capability node is HMAC-signed over its semantic fields; the store rejects tampered nodes at index time, providing supply-chain integrity. Resolution and execution emit chained, signed receipts, giving end-to-end provenance from intent to artifact. Trust levels gate which providers may satisfy an intent, and capability tokens with contextual caveats [Birgisson et al., 2014] bound authority. These mechanisms address several OWASP LLM risks [OWASP Foundation, 2025],

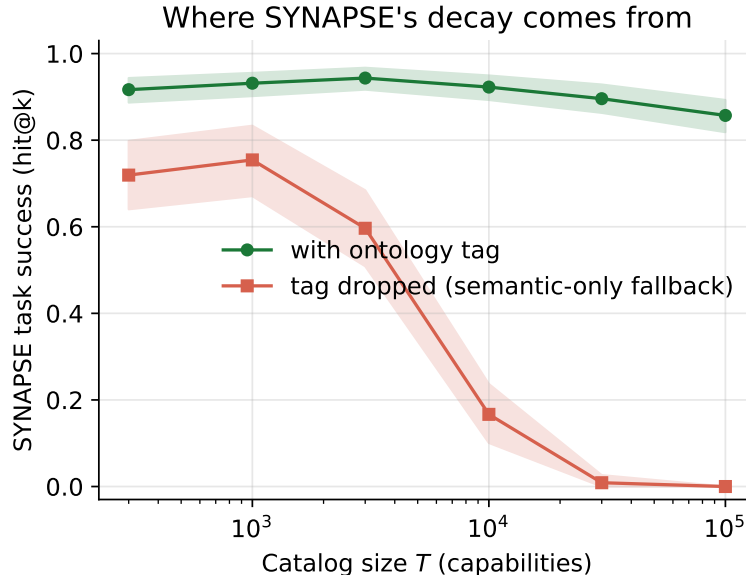


Figure 3: SYNAPSE success split by ontology-tag availability. The tagged subset stays near-flat across 2.5 orders of magnitude of catalog growth; the tag-dropout subset drives the mild aggregate decay.

Table 4: Resolver ablations at $T=10k$ (from `results/ablation.csv`).

Configuration	prec@1	hit@k	MRR
hybrid (full)	0.622	0.738	0.668
–semantic	0.000	0.000	0.000
–tag	0.260	0.491	0.326
–expand	0.600	0.711	0.643
–type-fit	0.602	0.724	0.652
–lexical	0.613	0.727	0.658
semantic-only	0.162	0.402	0.228

including excessive agency and supply-chain tampering, by making capability provenance explicit and verifiable rather than implicit in a prompt.

8 Reproducibility

The artifact is fully deterministic and offline. A single command (`python run_all.py`) regenerates every number, table, and figure from seed. Conformance tests encode the protocol’s normative requirements. We release the implementation, benchmark generators, analysis, and figure scripts.

9 Conclusion

The dominant tool protocols make agents *worse* as their ecosystems grow, because availability is conflated with in-context presence. SYNAPSE breaks that conflation with a signed capability knowledge graph, a calibrated hybrid resolver, and six fixed meta-verbs, achieving per-turn context

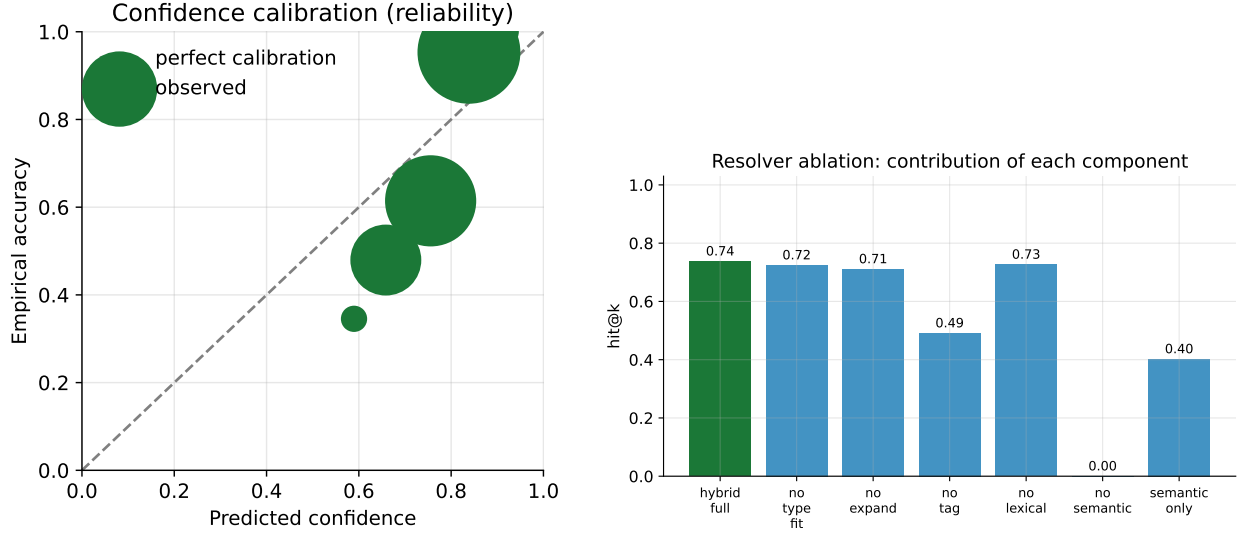


Figure 4: Left: confidence reliability (ECE = 0.138 pooled). Right: resolver ablation, where each signal contributes and the semantic and ontology-tag channels dominate.

and cost that are constant in the number of tools. Our matched-scorer evaluation shows this constancy is not merely cheaper but *prevents* the accuracy collapse that flat-context protocols suffer at scale, while we quantify SYNAPSE’s residual limitations honestly. SYNAPSE offers a practical, adoptable path, wrapping existing MCP servers as providers, to agent ecosystems that get *better*, not worse, as they grow.

A Notation

Table 5: Notation used throughout.

Symbol	Meaning
$\mathbf{C}, T$	Capability catalog and its size $ \mathbf{C} = T$.
k	Activation-set cap / baseline top- k (default 8).
W, T^*	MCP window (32,000 tokens) and its saturation size $W/\bar{L}$.
$\bar{L}$	Mean tool-description length in tokens (≈ 70.9).
N	Fixed number of gold evaluation tasks (90).
ϕ, sem	Embedder $\mathbb{R}^d$ and cosine similarity (Eq. (4)).
$\alpha, \beta, \gamma, \delta, \varepsilon$	Convex resolver weights (Eq. (5)).
θ	Relevance floor for grounded miss (Eq. (9)).
κ	Constant per-turn context bound (Theorem 1).
$\sqsubseteq, \sqcap, \sqcup, \top, \perp$	Subtype order, meet, join, top, bottom of $\mathbb{T}$.
Δ^+	Maximum typed out-degree of the CKG (Lemma 2).
σ, D	Seed breadth (40) and expansion depth (2).
λ	Proximity-kernel decay on graph distance (Eq. (10)).

B Extended Remarks on the Proofs

Tightness of Theorem 1. The bound $\kappa = B + 6s_{\max} + km$ is tight up to the tier-1 summary length: a full activation set of size k with maximal-length summaries attains κ , while a grounded miss (Eq. (9)) yields a strictly smaller context (only B plus the six verb schemas). The measured 31-token drift across 2.5 decades is activation sets filling toward k , not any dependence on T .

The visibility ceiling (Theorem 2). The argument is an expectation bound, not a curve fit. Pointwise $\text{hit}@k \leq \mathbf{1}[\text{gold visible}]$, and Wald’s identity [Wald, 1947] gives $\mathbb{E}[V] \bar{L} = W$, so $\mathbb{E}[V] \approx T^* = W/\bar{L}$ and $\Pr[\text{visible}] \approx \min(1, T^*/T)$. The half point lands at $T = 2T^*$ (Corollary 2), and the tail decays as $\Theta(1/T)$. The lone $T=3000$ point that grazes the ceiling does so within the 95% interval of five seeds, so the bound holds up to sampling error.

Bounded expansion (Lemma 2). The geometric bound in Δ^+ contains no factor of T , so with the shipped $\sigma = 40$ and $D = 2$ the candidate pool stays small; this is why the resolution plane in Table 1 never carries a factor of T . The proximity kernel of Eq. (10) weights candidates by λ raised to the graph distance, a discrete diffusion score in the sense of Kondor and Lafferty [Kondor and Lafferty, 2002] and a typed cousin of PageRank locality [Page et al., 1999].

Composition of guarantees. Theorems 1 and 5 compose: the planner operates over the activation set $A(q)$ (size $\leq k$), so plan construction never loads the full catalog into context. The end-to-end pipeline preserves the $O(1)$ context bound at every meta-verb, and every emitted plan is type-safe by Theorem 5.

References

- Anthropic. Model context protocol (mcp) specification. <https://modelcontextprotocol.io>, 2024. Open protocol for connecting AI assistants to tools and data.
- Arnar Birgisson, Joe Gibbs Politz, Úlfar Erlingsson, Ankur Taly, Michael Vrabie, and Mark Lentczner. Macaroons: Cookies with contextual caveats for decentralized authorization in the cloud. In *Network and Distributed System Security Symposium (NDSS)*, 2014.
- Brian A. Davey and Hilary A. Priestley. *Introduction to Lattices and Order*. Cambridge University Press, 2nd edition, 2002.
- Malik Ghallab, Dana Nau, and Paolo Traverso. Automated planning: Theory and practice. *Morgan Kaufmann*, 2004.
- Google. Agent2agent (a2a) protocol. <https://a2a-protocol.org>, 2024. Open protocol for interoperable communication between AI agents.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. *International Conference on Machine Learning (ICML)*, 2017.
- Risi Imre Kondor and John Lafferty. Diffusion kernels on graphs and other discrete structures. In *International Conference on Machine Learning (ICML)*, 2002.

- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics (TACL)*, 2024.
- Yu A. Malkov and Dmitry A. Yashunin. Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs. 2018.
- OWASP Foundation. Owasp top 10 for large language model applications. <https://owasp.org/www-project-top-10-for-large-language-model-applications/>, 2025.
- Lawrence Page, Sergey Brin, Rajeev Motwani, and Terry Winograd. The PageRank citation ranking: Bringing order to the web. Technical Report 1999-66, Stanford InfoLab, 1999.
- Shishir G. Patil, Tianjun Zhang, Xin Wang, and Joseph E. Gonzalez. Gorilla: Large language model connected with massive apis. *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Yujia Qin, Shihao Liang, Yining Ye, Kunlun Zhu, Lan Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang, Bill Qian, et al. Toolllm: Facilitating large language models to master 16000+ real-world apis. In *International Conference on Learning Representations (ICLR)*, 2024.
- Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Abraham Wald. *Sequential Analysis*. John Wiley & Sons, 1947.
- Kilian Weinberger, Anirban Dasgupta, John Langford, Alex Smola, and Josh Attenberg. Feature hashing for large scale multitask learning. In *International Conference on Machine Learning (ICML)*, 2009.